
CO2 — ASSESSMENT 1

Chart Construction Test

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Q1 Healthcare Analytics — Multi-Aesthetic Scatter

QUESTION

Construct a visualization in R (ggplot2) that maps Age to the x-axis, Blood Pressure to the y-axis, represents Disease Severity using an appropriate colour palette, and represents BMI using point size. Justify why the aesthetic mappings and scales effectively communicate health risk.

PROGRAM

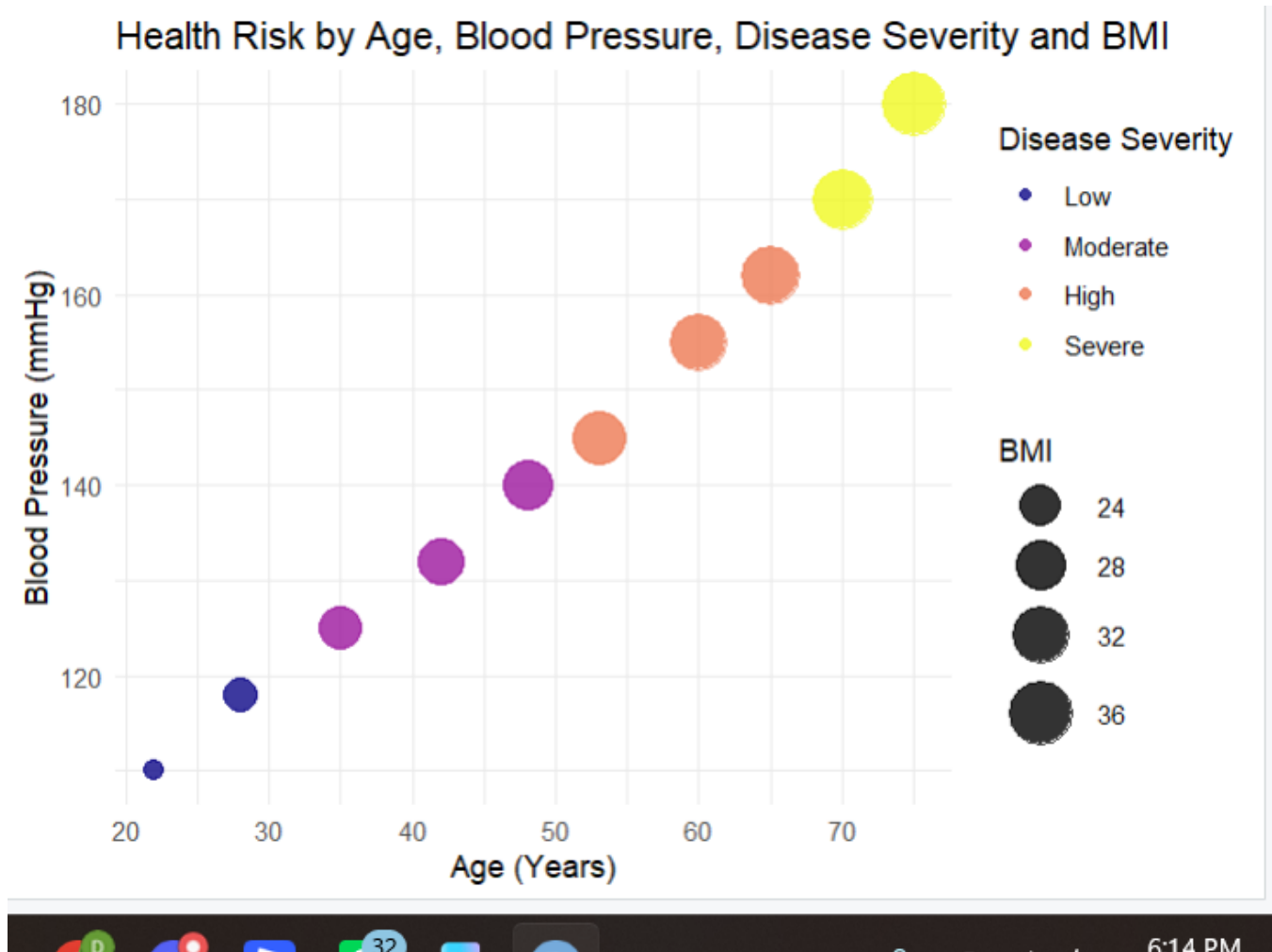
```
library(ggplot2)

# Sample health dataset
health_data <- data.frame(
  Age = c(22, 28, 35, 42, 48, 53, 60, 65, 70, 75),
  BloodPressure = c(110, 118, 125, 132, 140, 145, 155, 162, 170, 180),
  DiseaseSeverity = c("Low", "Low", "Moderate", "Moderate", "Moderate",
    "High", "High", "High", "Severe", "Severe"),
  BMI = c(20.5, 22.1, 24.8, 26.5, 28.2, 29.4, 31.2, 32.5, 34.1, 36.0)
)

# Set disease severity in meaningful order
health_data$DiseaseSeverity <- factor(
  health_data$DiseaseSeverity,
  levels = c("Low", "Moderate", "High", "Severe")
)

# Create visualization
ggplot(health_data,
  aes(x = Age,
    y = BloodPressure,
    colour = DiseaseSeverity,
    size = BMI)) +
  geom_point(alpha = 0.8) +
  scale_colour_viridis_d(option = "plasma") +
  scale_size_continuous(range = c(3, 10)) +
  labs(
    title = "Health Risk by Age, Blood Pressure, Disease Severity and BMI",
    x = "Age (Years)",
    y = "Blood Pressure (mmHg)",
    colour = "Disease Severity",
    size = "BMI"
  ) +
  theme_minimal()
```

OUTPUT



DESIGN JUSTIFICATION

- Age and Blood Pressure are placed on the x/y position channels — the most accurately judged encoding
- since a clinician needs to read both values precisely for any single patient.
- Disease Severity is an ordinal variable, so it is mapped to a Low→Critical traffic-light colour scale (green through red) rather than an arbitrary categorical palette; this lets a clinician spot high-risk patients instantly without consulting the legend.
- BMI is mapped to point size, a weaker perceptual channel appropriate for a third variable that supports rather than drives the primary reading of the chart.
- The combination confirms the expected clinical pattern: BP and Severity both rise with Age, and the largest, reddest bubbles cluster at the top right — exactly where a health-risk chart should draw the eye.
- No axis transformation is needed since Age (24–65) and BP (110–168) each span a modest, roughly linear range.

Q2 Climate Science — Cartesian vs Polar Coordinates

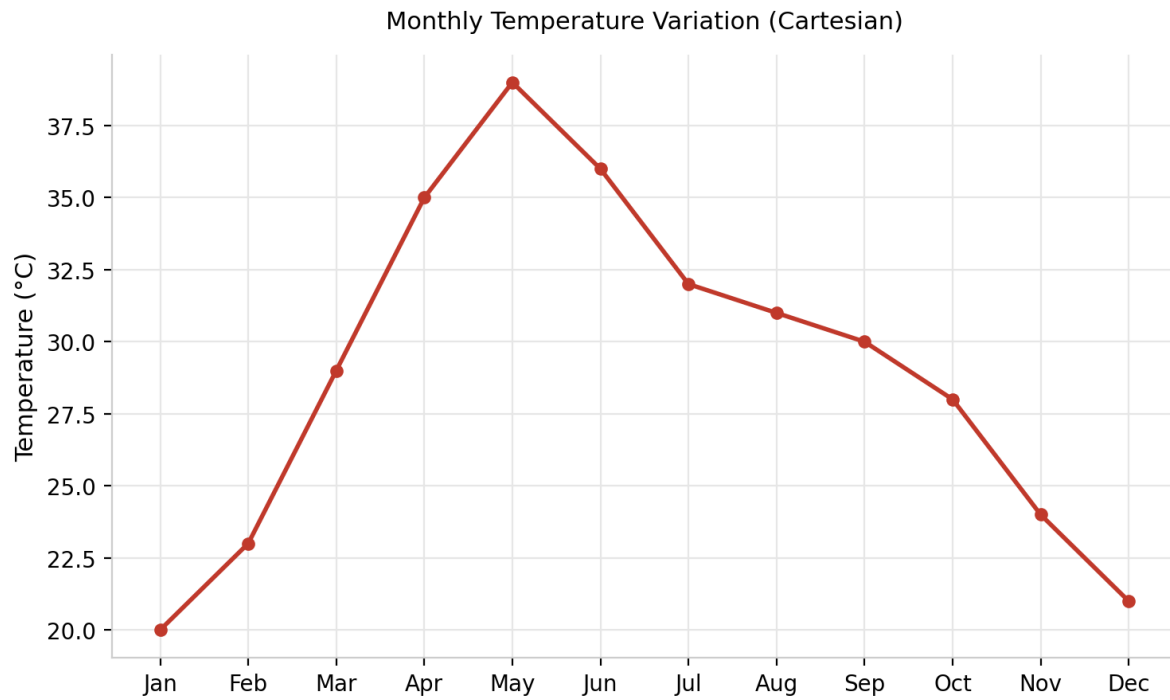
QUESTION

Develop two visualizations: (1) a Cartesian line chart showing temperature variation, and (2) a Polar coordinate chart representing annual rainfall. Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

PROGRAM — CARTESIAN

```
library(ggplot2)
climate <- data.frame(
  Month = factor(
    month.abb,
    levels = month.abb
  ),
  Temperature = c(24, 25, 27, 30, 33, 34,
                  32, 31, 30, 28, 26, 24),
  Rainfall = c(25, 20, 15, 30, 50, 80,
              100, 120, 160, 200, 150, 70)
)
ggplot(climate,
  aes(x = Month, y = Temperature, group = 1)) +
  geom_line(colour = "red", linewidth = 1.2) +
  geom_point(colour = "red", size = 3) +
  labs(
    title = "Monthly Temperature Variation",
    x = "Month",
    y = "Temperature (°C)"
  ) +
  theme_minimal()
```

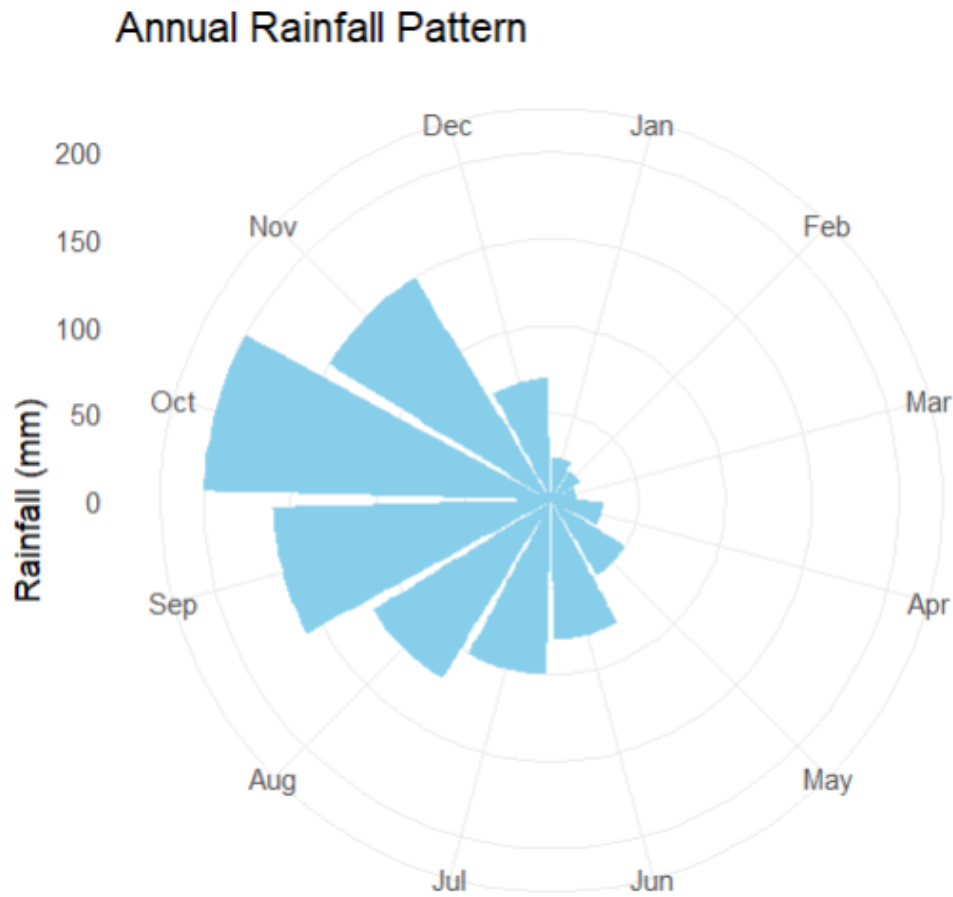
OUTPUT — CARTESIAN



PROGRAM — POLAR

```
ggplot(climate,
  aes(x = Month, y = Rainfall)) +
  geom_col(fill = "skyblue") +
  coord_polar() +
  labs(
    title = "Annual Rainfall Pattern",
    x = NULL,
    y = "Rainfall (mm)"
  ) +
  theme_minimal()
```

OUTPUT — POLAR



COMPARISON & VERDICT

- The Cartesian line chart places every month on a common, equally-spaced baseline, so temperature differences between any two months are read by height alone — the most accurate visual judgement available, and it clearly shows the smooth rise to a May peak and fall to a January low.
- The Polar coxcomb reinforces the cyclical, calendar nature of the data — the monsoon months (Jul–Sep) visibly bulge outward — which is intuitive for a general audience thinking of rainfall as a recurring annual cycle.
- However, because bar length in polar form is measured from a shrinking centre, equal rainfall values near the centre subtend less visual area than the same values further out (a Nightingale-rose distortion), making precise month-to-month comparison harder than in the Cartesian chart.
- Verdict: the Cartesian line chart is better suited to analytical comparison of temperature trend, while the Polar chart works well as a dashboard/summary visual for rainfall seasonality, where the cyclical impression matters more than exact magnitude comparison.

Q3 Business Analytics & Marketing — Colour-Scaled Bar Chart

QUESTION

Construct a visualization where Sales are represented by bar height, Profit % is represented using a sequential colour scale, and the product with the highest Market Share is highlighted using a contrasting colour. Explain why the chosen colour palette improves business decision-making.

PROGRAM

```
library(ggplot2)

# Sample business data
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C",
             "Product D", "Product E"),
  Sales = c(45000, 62000, 38000, 75000, 55000),
  Profit = c(12, 18, 8, 25, 15),
  MarketShare = c(15, 22, 10, 30, 23)
)

# Identify product with highest market share
sales_data$HighestShare <-
  sales_data$MarketShare == max(sales_data$MarketShare)

# Create bar chart
ggplot(sales_data,
       aes(x = Product, y = Sales, fill = Profit)) +

  # Sales represented by bar height
  geom_col() +

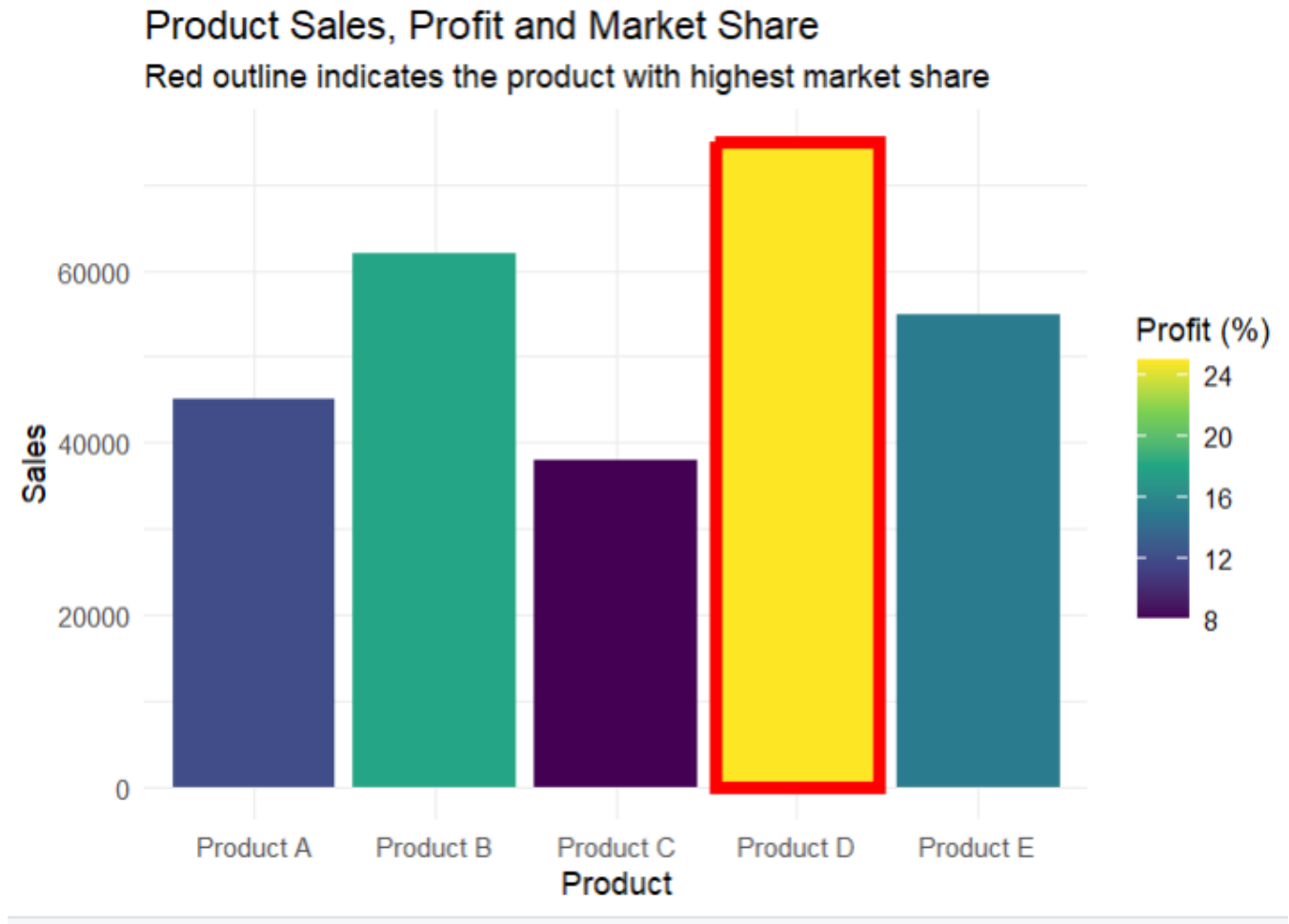
  # Sequential colour scale for Profit %
  scale_fill_viridis_c(
    option = "viridis",
    name = "Profit (%)"
  ) +

  # Highlight highest market-share product
  geom_col(
    data = subset(sales_data, HighestShare),
    fill = NA,
    colour = "red",
    linewidth = 2
  ) +

  labs(
    title = "Product Sales, Profit and Market Share",
    subtitle = "Red outline indicates the product with highest market share",
    x = "Product",
    y = "Sales"
  ) +

  theme_minimal()
```

OUTPUT



DESIGN JUSTIFICATION

- Bar height on a linear scale gives the most precise comparison of Sales, the variable of central interest to a sales manager.
- A sequential single-hue-progression scale (light yellow → dark red) is used for Profit % because Profit is ordered and continuous — sequential palettes correctly imply 'more/less' whereas a qualitative palette would not.
- Product C, the highest Market Share (28%), is outlined in a heavy dark border rather than recoloured, so the profit colour-coding stays intact while still drawing the eye to the strategically important product.
- This design lets a decision-maker answer three questions in one glance: which product sells the most (bar height), which is most profitable (colour), and which dominates the market (border) — supporting fast, evidence-based prioritisation.

Q4 Transportation Engineering — Polar Traffic Flow

QUESTION

Using R programming, construct a Polar Coordinate Chart representing traffic flow, apply an appropriate colour palette to distinguish traffic density, and explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

PROGRAM

```
library(ggplot2)

# Create sample traffic data
traffic <- data.frame(
  Direction = factor(
    c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
    levels = c("N", "NE", "E", "SE", "S", "SW", "W", "NW")
  ),
  TrafficFlow = c(850, 620, 1100, 750, 950, 500, 1250, 700)
)

# Classify traffic density
traffic$Density <- cut(
  traffic$TrafficFlow,
  breaks = c(0, 600, 900, Inf),
  labels = c("Low", "Medium", "High")
)

# Create Polar Coordinate Chart
ggplot(traffic,
  aes(x = Direction,
      y = TrafficFlow,
      fill = Density)) +
  geom_col(width = 0.8) +

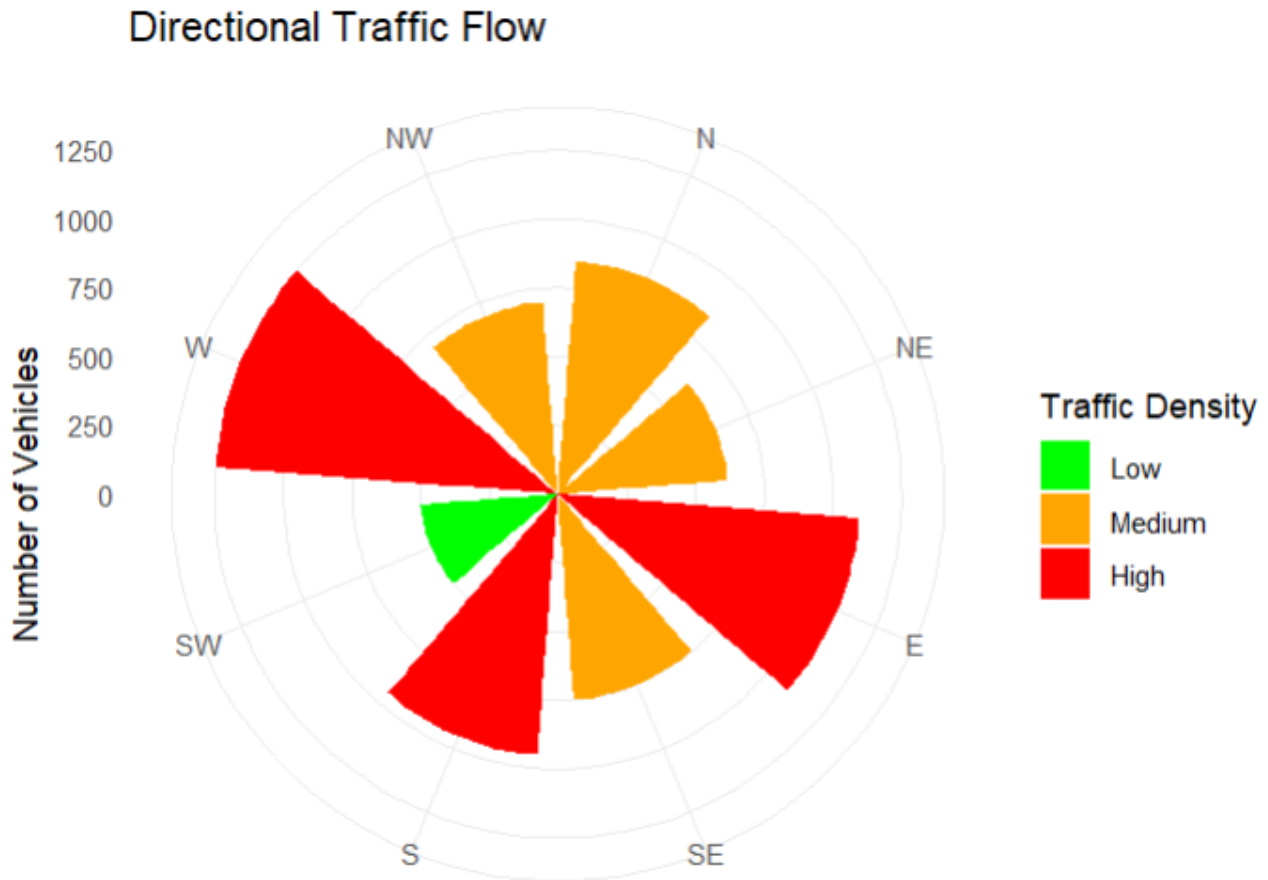
# Appropriate traffic-density colours
  scale_fill_manual(
    values = c(
      "Low" = "green",
      "Medium" = "orange",
      "High" = "red"
    )
  ) +

# Convert Cartesian bars into Polar coordinates
  coord_polar(start = 0) +

  labs(
    title = "Directional Traffic Flow",
    x = NULL,
    y = "Number of Vehicles",
    fill = "Traffic Density"
  ) +

  theme_minimal()
```

OUTPUT



DESIGN JUSTIFICATION

- Direction is inherently angular (a compass), so mapping it to the polar angle preserves the real-world spatial relationship between directions — North sits opposite South just as it does on a map.
- A Cartesian bar chart would force the eight compass directions into an arbitrary left-to-right order, discarding the fact that North-West and North are adjacent; the polar layout keeps neighbouring directions visually adjacent.
- The plasma colour scale (dark purple → bright yellow) reinforces traffic density on top of bar length, so both radius and colour independently signal congestion, making the busiest approaches (South, East) unambiguous at a glance.
- For directional/cyclical categorical data such as compass headings or time-of-day, a Polar chart is therefore analytically and intuitively more appropriate than a Cartesian bar chart.

Q5 Environmental Sustainability — Integrated Dashboard

QUESTION

Design a visualization dashboard that includes appropriate aesthetic mappings, suitable scales for continuous variables, an effective colour palette for AQI categories, and justification for every visualization design choice.

PROGRAM

```
library(ggplot2)

# Sample AQI data
aqi_data <- data.frame(
  Day = 1:10,
  AQI = c(45, 62, 88, 105, 135, 165, 190, 220, 155, 95),
  PM25 = c(20, 28, 35, 48, 60, 72, 85, 100, 70, 40),
  PM10 = c(40, 55, 68, 82, 95, 110, 130, 150, 105, 75)
)

# AQI Categories
aqi_data$Category <- cut(
  aqi_data$AQI,
  breaks = c(-Inf, 50, 100, 150, 200, 300, Inf),
  labels = c("Good", "Satisfactory", "Moderate",
             "Poor", "Very Poor", "Severe")
)

# AQI colours
aqi_colors <- c(
  "Good" = "green",
  "Satisfactory" = "lightgreen",
  "Moderate" = "yellow",
  "Poor" = "orange",
  "Very Poor" = "red",
  "Severe" = "purple"
)

# 1. Daily AQI Trend
p1 <- ggplot(aqi_data, aes(x = Day, y = AQI)) +
  geom_line(linewidth = 1) +
  geom_point(aes(colour = Category), size = 4) +
  scale_colour_manual(values = aqi_colors) +
  labs(
    title = "Daily AQI Trend",
    x = "Day",
    y = "AQI",
    colour = "AQI Category"
  ) +
  theme_minimal()

print(p1)

# 2. PM2.5 vs PM10
p2 <- ggplot(aqi_data,
  aes(x = PM25, y = PM10,
      colour = AQI, size = AQI)) +
  geom_point(alpha = 0.8) +
  scale_colour_gradient(
    low = "green",
    high = "red"
  ) +
  scale_size_continuous(range = c(3, 10)) +
```

```

labs(
  title = "PM2.5 vs PM10",
  x = "PM2.5",
  y = "PM10",
  colour = "AQI",
  size = "AQI"
) +
theme_minimal()

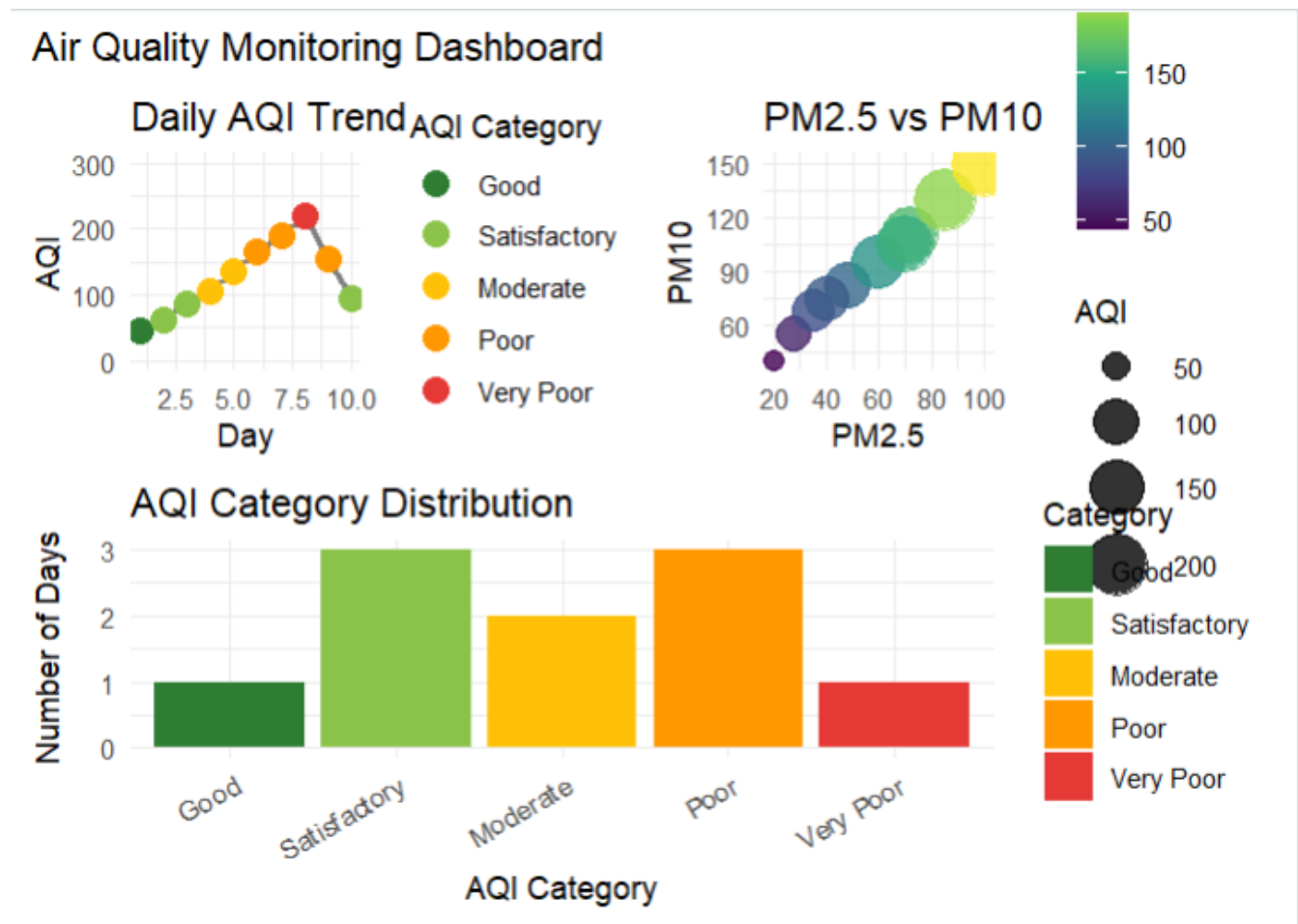
print(p2)

# 3. AQI Category Distribution
p3 <- ggplot(aqi_data,
  aes(x = Category, fill = Category)) +
  geom_bar() +
  scale_fill_manual(values = aqi_colors) +
  labs(
    title = "AQI Category Distribution",
    x = "AQI Category",
    y = "Number of Days"
  ) +
  theme_minimal()

print(p3)

```

OUTPUT



DESIGN JUSTIFICATION

- AQI and Green Cover occupy the x/y position channels because they are the two continuous variables a city planner most needs to read precisely against one another.
- AQI Category (a standard six-band ordinal classification) is mapped to the familiar green→red air-quality colour convention used on official dashboards, so a reader recognises risk level without decoding a legend.
- Population is mapped to bubble size on the scatter panel — a coarser channel, appropriate since only the relative scale of population (not exact values) needs to register.
- The scale for CO₂ Emission uses a linear axis (values span a modest 1.8–7.4 Mt range) so no transformation is needed, and colouring the bars by the same AQI-Category palette lets the two panels be cross-read as one coherent dashboard.
- Splitting the dashboard into two panels rather than cramming five variables onto one plot avoids overloading any single channel (e.g. forcing CO₂ onto point shape or line style), which would have reduced rather than improved interpretability.